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```
clear;
clc;
close all;
```

Load Talley's mixed spark + noise file

```
data = load('SPARK_NOISE_EQUAL_RMS_001.mat');

iq = data.eventIQ;
fs = data.fs;
fc = data.fc;

fprintf('Sample rate: %.0f Hz\n', fs);
fprintf('Center frequency: %.1f MHz\n\n', fc/1e6);

Sample rate: 2400000 Hz
Center frequency: 145.9 MHz
```

Magnitude

```
mag = abs(iq);
```

Smooth the signal

```
windowLength = 200;
smoothMag = movmean(mag, windowLength);
```

Estimate background level

```
baseline = median(smoothMag);
```

Detection settings

```
threshold = baseline * 1.5;
```

```
% Main fix from v04:  
% require events to be at least 6 ms apart  
minPeakDistance = round(0.006 * fs);  
  
% Require peaks to stand out from local background  
minProminence = 0.006;  
  
fprintf('Baseline: %.6f\n', baseline);  
fprintf('Threshold: %.6f\n', threshold);  
fprintf('Minimum event spacing: 6 ms\n');  
fprintf('Minimum prominence: %.6f\n\n', minProminence);
```

```
Baseline: 0.026469  
Threshold: 0.039703  
Minimum event spacing: 6 ms  
Minimum prominence: 0.006000
```

Detect candidate events

```
[peakValues, peakLocations] = findpeaks( ...  
    smoothMag, ...  
    'MinPeakHeight', threshold, ...  
    'MinPeakProminence', minProminence, ...  
    'MinPeakDistance', minPeakDistance);
```

Convert sample locations to time

```
peakTimes = peakLocations / fs;
```

Print events

```
fprintf('===== DETECTED EVENTS =====\n');  
fprintf('Number of candidate events: %d\n\n', length(peakTimes));  
  
for k = 1:length(peakTimes)  
    fprintf('Event %d: %.6f s\n', k, peakTimes(k));  
end
```

```
===== DETECTED EVENTS =====  
Number of candidate events: 120
```

Event 1: 0.002928 s
Event 2: 0.011260 s
Event 3: 0.019654 s
Event 4: 0.027928 s
Event 5: 0.036260 s
Event 6: 0.044587 s
Event 7: 0.052996 s
Event 8: 0.061323 s
Event 9: 0.069590 s
Event 10: 0.077920 s
Event 11: 0.086264 s
Event 12: 0.094641 s
Event 13: 0.102927 s
Event 14: 0.111262 s
Event 15: 0.119595 s
Event 16: 0.127934 s
Event 17: 0.136192 s
Event 18: 0.144598 s
Event 19: 0.152977 s
Event 20: 0.161259 s
Event 21: 0.169645 s
Event 22: 0.177863 s
Event 23: 0.186263 s
Event 24: 0.194658 s
Event 25: 0.202925 s
Event 26: 0.211315 s
Event 27: 0.219530 s
Event 28: 0.227858 s
Event 29: 0.236256 s
Event 30: 0.244655 s
Event 31: 0.252975 s
Event 32: 0.261194 s
Event 33: 0.269590 s
Event 34: 0.277969 s
Event 35: 0.286264 s
Event 36: 0.294527 s
Event 37: 0.302862 s
Event 38: 0.311193 s
Event 39: 0.319660 s
Event 40: 0.327990 s
Event 41: 0.336258 s
Event 42: 0.344594 s
Event 43: 0.352935 s
Event 44: 0.361303 s
Event 45: 0.369454 s
Event 46: 0.377918 s
Event 47: 0.386259 s
Event 48: 0.394595 s
Event 49: 0.402972 s
Event 50: 0.411197 s
Event 51: 0.419532 s
Event 52: 0.427922 s
Event 53: 0.436307 s
Event 54: 0.444652 s

Event 55: 0.452979 s
Event 56: 0.461261 s
Event 57: 0.469653 s
Event 58: 0.477918 s
Event 59: 0.486267 s
Event 60: 0.494660 s
Event 61: 0.502931 s
Event 62: 0.511199 s
Event 63: 0.519593 s
Event 64: 0.527971 s
Event 65: 0.536197 s
Event 66: 0.544591 s
Event 67: 0.552987 s
Event 68: 0.561325 s
Event 69: 0.569530 s
Event 70: 0.577928 s
Event 71: 0.586121 s
Event 72: 0.594527 s
Event 73: 0.602923 s
Event 74: 0.611318 s
Event 75: 0.619525 s
Event 76: 0.627923 s
Event 77: 0.636329 s
Event 78: 0.644599 s
Event 79: 0.652863 s
Event 80: 0.661266 s
Event 81: 0.669532 s
Event 82: 0.677979 s
Event 83: 0.686313 s
Event 84: 0.694585 s
Event 85: 0.702972 s
Event 86: 0.711194 s
Event 87: 0.719595 s
Event 88: 0.727860 s
Event 89: 0.736318 s
Event 90: 0.744657 s
Event 91: 0.752861 s
Event 92: 0.761262 s
Event 93: 0.769638 s
Event 94: 0.777985 s
Event 95: 0.786197 s
Event 96: 0.794661 s
Event 97: 0.802866 s
Event 98: 0.811199 s
Event 99: 0.819530 s
Event 100: 0.827927 s
Event 101: 0.836193 s
Event 102: 0.844638 s
Event 103: 0.852918 s
Event 104: 0.861263 s
Event 105: 0.869599 s
Event 106: 0.877866 s
Event 107: 0.886261 s
Event 108: 0.894454 s

```
Event 109: 0.902931 s
Event 110: 0.911321 s
Event 111: 0.919530 s
Event 112: 0.927924 s
Event 113: 0.936256 s
Event 114: 0.944526 s
Event 115: 0.952990 s
Event 116: 0.961261 s
Event 117: 0.969654 s
Event 118: 0.977920 s
Event 119: 0.986198 s
Event 120: 0.994527 s
```

Event spacing

```
if length(peakTimes) > 1

    spacing = diff(peakTimes);

    avgSpacing = mean(spacing);
    medianSpacing = median(spacing);

    fprintf('\n===== EVENT SPACING =====\n');
    fprintf('Average spacing: %.3f ms\n', avgSpacing * 1000);
    fprintf('Median spacing: %.3f ms\n', medianSpacing * 1000);
```

```
===== EVENT SPACING =====
Average spacing: 8.333 ms
Median spacing: 8.336 ms
```

Estimate repetition rate

```
repetitionRate = 1 / medianSpacing;

fprintf('Estimated repetition rate: %.2f Hz\n', repetitionRate);

Estimated repetition rate: 119.96 Hz
```

Compare with expected 120 Hz-like behavior

```
expectedSpacing = 1 / 120;

spacingError = abs(medianSpacing - expectedSpacing);

fprintf('Expected 120 Hz spacing: %.3f ms\n', ...
    expectedSpacing * 1000);

fprintf('Spacing error: %.3f ms\n', ...
    spacingError * 1000);

Expected 120 Hz spacing: 8.333 ms
Spacing error: 0.002 ms
```

end

Plot first 100 ms

```
showTime = 0.1;
Nshow = round(showTime * fs);

t = (0:length(iq)-1) / fs;

figure;

plot(t(1:Nshow)*1000, smoothMag(1:Nshow));
hold on;

yline(threshold, '--', 'Detection Threshold');

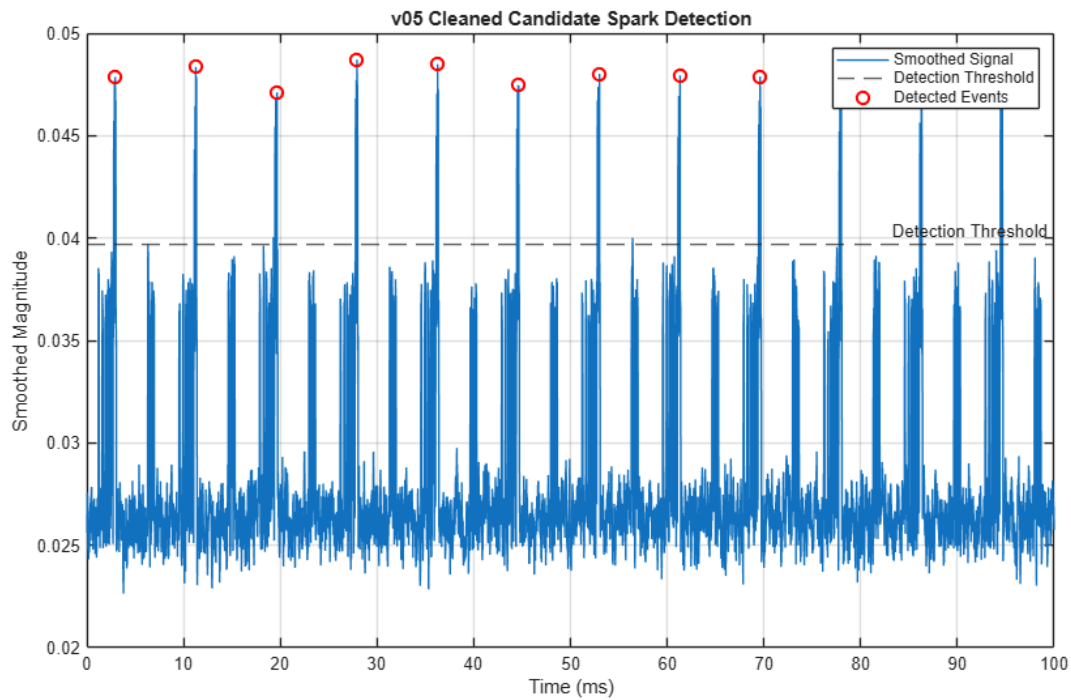
visibleEvents = peakLocations <= Nshow;

plot( ...
    peakTimes(visibleEvents)*1000, ...
    peakValues(visibleEvents), ...
    'ro', ...
    'MarkerSize', 7, ...
    'LineWidth', 1.5);

xlabel('Time (ms)');
ylabel('Smoothed Magnitude');
title('v05 Cleaned Candidate Spark Detection');
grid on;

legend( ...
    'Smoothed Signal', ...
    'Detection Threshold', ...
    'Detected Events');

hold off;
```



Final detector summary

```
fprintf('\n===== DETECTOR SUMMARY =====\n');

if isempty(peakTimes)

    fprintf('NO CANDIDATE SPARK EVENTS DETECTED\n');

else

    fprintf('CANDIDATE SPARK EVENTS DETECTED\n');

    if length(peakTimes) > 1

        if medianSpacing > 0.007 && medianSpacing < 0.010
            fprintf('REPEATING EVENT PATTERN DETECTED\n');
        else
            fprintf('EVENTS DETECTED, BUT TIMING PATTERN IS NOT
CONSISTENT\n');
        end

    end

end

===== DETECTOR SUMMARY =====
```

CANDIDATE SPARK EVENTS DETECTED
REPEATING EVENT PATTERN DETECTED

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